FINDINGS OF COHORT STUDY ON POLYTECHNIC GRADUATES ACQUIRING UNIVERSITY LEVEL EDUCATION

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This paper is a collaborative effort between
Manpower Planning & Policy Division, Ministry of Manpower and
Higher Education Division, Ministry of Education.

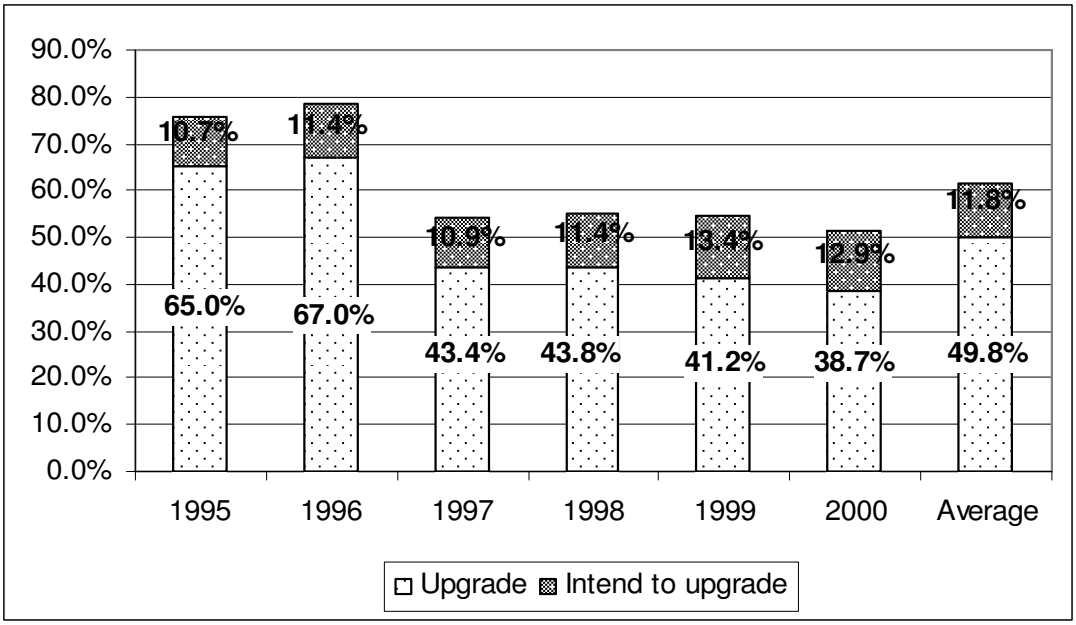
Introduction

1. Anecdotally, there have been observations that more polytechnic graduates were pursuing university education in recent years. To better understand this upgrading trend and its returns, the Ministries of Education (MOE) and Manpower (MOM) conducted surveys[1] on polytechnic graduate cohorts from 1995 to 2000.

Survey Findings

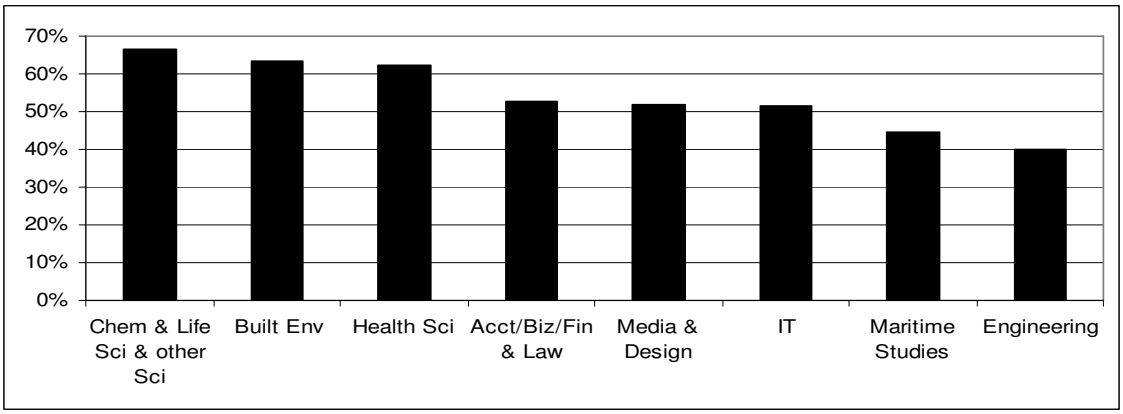
2. **Likelihood of upgrading.** The results showed that about 4 in 10 polytechnic graduates can be expected to upgrade themselves and obtain a degree within 5 years of graduation (see Chart 1). Amongst the various polytechnic courses, graduates from *Chemical & Life Sciences & Other Sciences*, *Built Environment*, and *Health Sciences* were the most likely to upgrade, while *Engineering* graduates were the least likely (see Chart 2).

Chart 1: Proportion of Polytechnic Upgraders by Graduating Cohort



[1] The surveys, conducted in 2006, covered a total of 73,365 polytechnic graduates from Singapore Polytechnic, Ngee Ann Polytechnic, Temasek Polytechnic and Nanyang Polytechnic. 22,670 of them responded to the surveys, equivalent to a response rate of 30.9%.

Chart 2: Proportion who Upgraded within each Polytechnic Course (1995-2000)



3. **Length and Type of Degree Programme.** Most polytechnic upgraders undertook degree courses with durations between 13 and 36 months (see Chart 3). In terms of type of degree programmes chosen, the majority of the upgraders pursued their further studies by going abroad or taking distance learning programmes, i.e., foreign degree programmes offered locally (see Chart 4).

Chart 3: Distribution of Upgraders by Length of University Course (1995-2000)

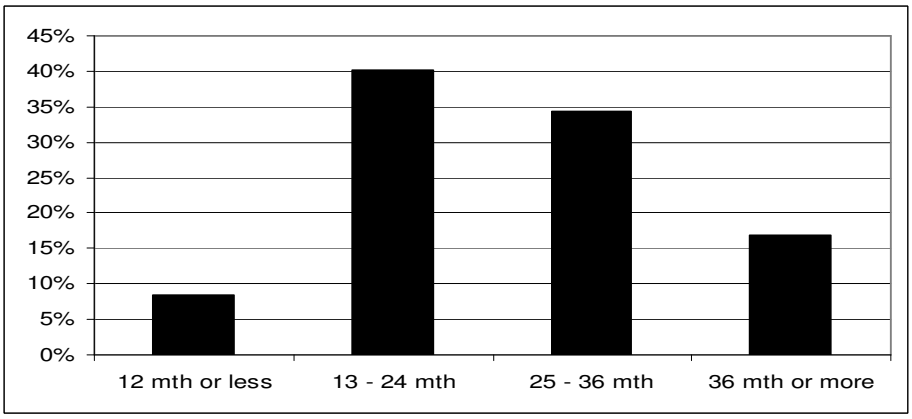
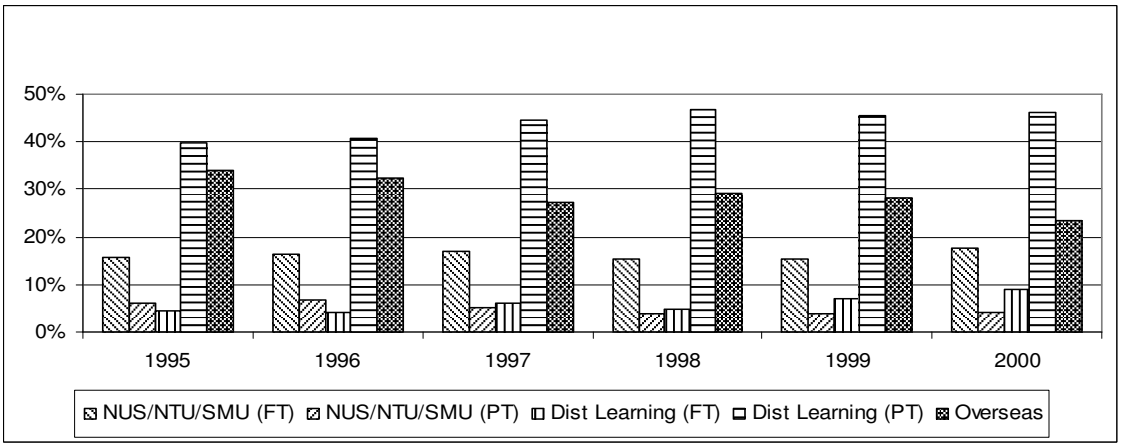


Chart 4: Distribution of Upgraders by Types of Upgrading (1995-2000)



Analysis of Results

4. **Cohort Education Profile.** Based on the long-term polytechnic target cohort participation rate (CPR) of 40%, within each local student cohort, about 4 in 10 will be admitted to the polytechnics. As shown from the survey, amongst them, 5 in 10 are expected to upgrade to degrees within 5 to 10 years of obtaining their diplomas. This upgrading will increase the total proportion of degree holders in each cohort from 25%[2] to about 40%[3], and correspondingly decrease the share of the cohort with diplomas.

5. **Choices of Courses.** The survey results also suggest that at least three quarters of upgraders from all S&T[4] polytechnic courses chose to continue pursuing S&T education in university (see Table 1). More than half of upgraders from S&T polytechnic courses continued to be employed in S&T occupations within 5 to 10 years of graduation, slightly higher than the corresponding proportion for non-upgraders (see Table 2).

Table 1: University Courses Studied by Polytechnic Upgraders

University Course \ Polytechnic Course		S&T				Non-S&T			
		Engineering	Chem & Life Sci & other Sci	Health Sci	IT	Built Env	Maritime Studies	Acct/Biz/Fin & Law	Media & Design
% upgraded in S&T course		**73%**	**73%**	**92%**	**82%**	**30%**	**51%**	**3%**	**3%**
S&T	**Engineering**	54%	16%	0%	2%	19%	42%	1%	0%
	Natural and Phy Sci	2%	21%	0%	0%	1%	2%	1%	0%
	Health/Life/Bio Sci	1%	34%	90%	1%	7%	2%	0%	0%
	IT	16%	2%	1%	80%	2%	5%	2%	2%
Non-S&T	**Media & Design**	1%	0%	0%	1%	0%	1%	1%	29%
	Biz/Fin/Mgt	17%	13%	3%	8%	14%	28%	63%	9%
	Law/Acc	1%	0%	1%	1%	2%	2%	19%	0%
	Social Sci	1%	1%	3%	1%	1%	1%	2%	5%
	Arts	0%	0%	0%	0%	0%	0%	0%	4%
	Tourism	0%	0%	0%	0%	0%	0%	1%	0%
	Arch/ Real Estate	0%	0%	0%	0%	31%	1%	0%	2%
	Others	6%	11%	1%	6%	21%	16%	11%	47%

Note:

(i) The percentages in each column show the distribution of university courses studied by upgraders from a particular polytechnic course.

(ii) Figures have been calculated based on all cohorts from 1995-2000. Totals may not tally due to rounding.

[2] Assuming an estimated 3% of each student cohort who were 'A' level holders getting degrees from studying overseas, and university and polytechnic CPRs of 22% and 38% respectively.

[3] Assuming that 45% of polytechnic graduates upgrade to become degree holders.

[4] Science & Technology (S&T) in this paper refers to Engineering, Information Technology and the Sciences (including Health Science).

Table 2: Proportion of Upgraders and Non-Upgraders in S&T Occupations

Course Type	Polytechnic Course	% in S&T Occupations	
		Upgraders	Non-upgraders
S&T	Engineering	54%	42%
	Chem & Life Sci & other Sci	46%	52%
	Health Sci	78%	78%
	IT	56%	32%
	All S&T Courses	**56%**	**46%**
Non-S&T	Built Env	37%	35%
	Maritime Studies	29%	23%
	Acct/Biz/Fin & Law	2%	2%
	Media & Design	5%	9%

Note:

(i) The values in each row show the percentage of graduates (both upgraders and non-upgraders) from a particular polytechnic course who are currently in S&T jobs. For example, the first row shows that 54% of upgraders from *Engineering* in polytechnic are currently in S&T jobs, while only 42% of those who studied Engineering in polytechnic but did not upgrade are currently in S&T jobs.

(ii) Figures have been calculated based on all cohorts from 1995-2000.

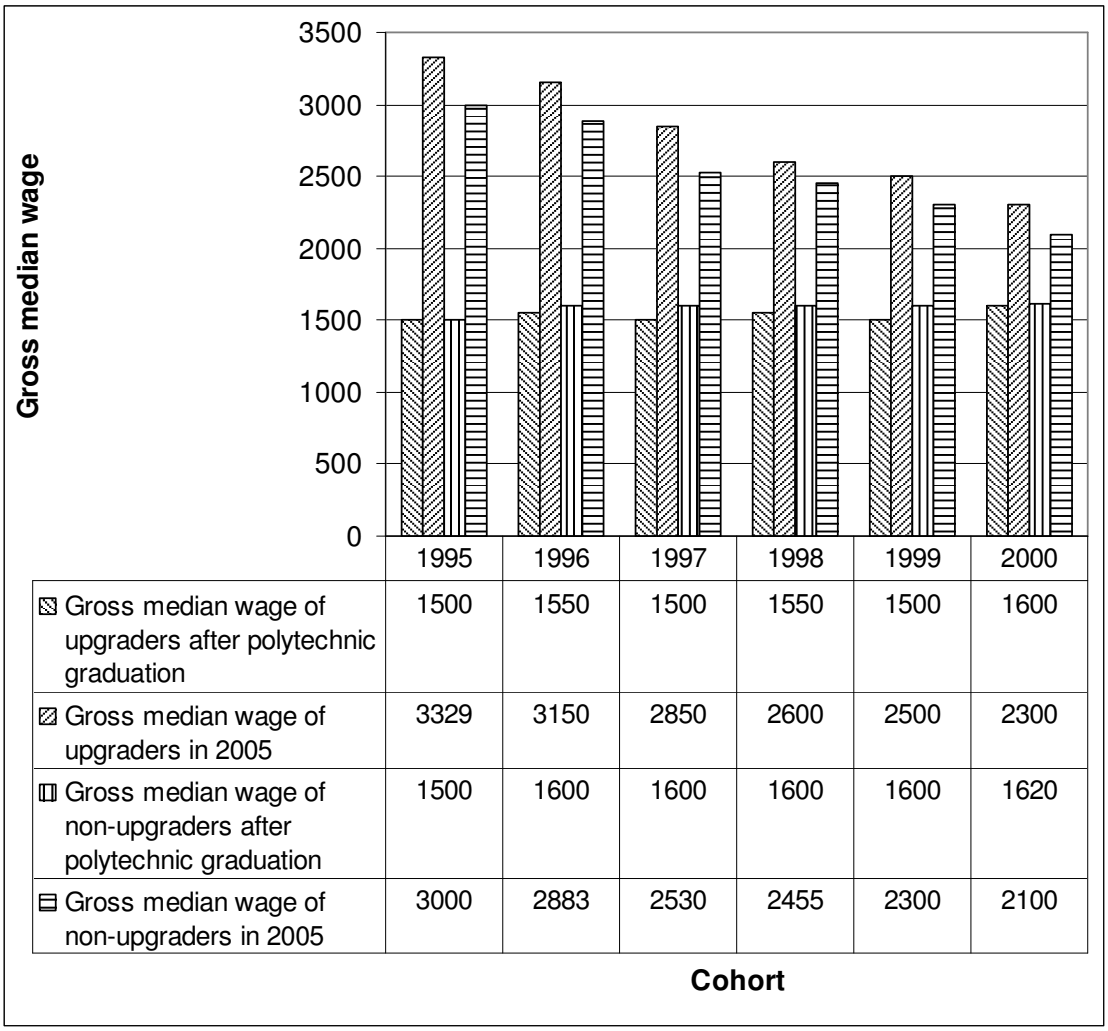
6. **Wages.** Further regression analysis was done to study the impact on the wages of polytechnic graduates who went on to obtain a degree. The wage premium for upgrading, after controlling for gender, work experience and cohort-specific effects[5], is estimated to be about 13%. Also, each additional year of work experience after polytechnic graduation increases the wage by 3% (see Chart 5).

9. The wage analysis revealed that upgraders had higher wage increases when they completed their university programmes in 3 to 4 years relative to a shorter programme of 1 to 2 years, suggesting that skills and human capital take time to acquire. After controlling for the length of the programme, no clear impact by the type of university degree[6] acquired on the wage premium was found. However, this might be due to the survey covering a relatively short time span (5 to 10 years after graduation from polytechnic), which does not allow for any significant difference to show up in the wages of polytechnic upgraders who pursued different university degrees.

[5] For example, there might be differences in the curriculum for a course for different cohorts.

[6] Degree can be obtained either at a local university, overseas university or through distance-learning.

Chart 5: Wages of Graduates by upgrade, wage after polytechnic, current wage and cohort



	1995	1996	1997	1998	1999	2000
Gross median wage of upgraders after polytechnic graduation	1500	1550	1500	1550	1500	1600
Gross median wage of upgraders in 2005	3329	3150	2850	2600	2500	2300
Gross median wage of non-upgraders after polytechnic graduation	1500	1600	1600	1600	1600	1620
Gross median wage of non-upgraders in 2005	3000	2883	2530	2455	2300	2100

Conclusion

10. The survey results showed that a substantial proportion of polytechnic graduates pursued university education within 5 to 10 years of graduation, with positive returns in the form of improved wages.